

Table 1 Geometric Features

No.	Feature Name	Physical meaning	Calculation Formula
1	obj_Area	Total pixel area of the target region, indicating the overall size of the target	$Area = \sum_{i \in \Omega} 1$
2	obj_Perimeter	Target contour perimeter, representing the length of the target's boundary	$Perimeter = L$
3	obj_AspectRatio	The aspect ratio of the bounding rectangle around the target characterizes the target's elongated shape	$AspectRatio = \frac{w}{h}$
4	obj_ShapeIndex	Shape index, which characterizes the degree of deviation between the target contour and a standard circle	$Shape\ Index = \frac{Perimeter}{\sqrt{2\pi \cdot Area}}$
5	obj_Circularity	Roundness: The closer the value is to 1, the closer the target contour is to a perfect circle.	$Circularity = \frac{4\pi \cdot Area}{Perimeter^2}$
6	obj_Compactness	Compactness, a measure of how compact the target contour is	$Compactness = \frac{Perimeter^2}{Area}$
7	obj_FourierEnergy _y	Contour Fourier descriptor energy, which characterizes the global contour shape features of the object	$FourierEnergy = \frac{1}{5} \sum_{k=1}^5 F_k $
8	pix_gx	The first-order gradient in the X direction of the image reflects the intensity of horizontal edges	$g_x = \frac{\partial I(x,y)}{\partial x}$
9	pix_gy	The first-order gradient in the Y direction of the image reflects the intensity of vertical edges	$g_y = \frac{\partial I(x,y)}{\partial y}$
10	pix_grad_mag	Gradient magnitude, a comprehensive measure of edge strength in a local region	$GradMag = \sqrt{g_x^2 + g_y^2}$
11	pix_grad_ori	Gradient direction, representing the spatial orientation angle of local edges	$GradOri = \arctan 2(g_y, g_x)$
12	pix_laplacian	Characterizing second-order edges, corner points, and regions of sudden changes in grayscale	$Lap = \frac{\partial^2 I(x,y)}{\partial x^2} + \frac{\partial^2 I(x,y)}{\partial y^2}$
13	pix_hessian_l1	The first eigenvalue of a matrix (the trace of the matrix) characterizes local surface curvature	$H_1 = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$
14	pix_hessian_l2	The second eigenvalue of the matrix, which characterizes local morphological differences	$H_2 = \frac{\partial^2 I}{\partial x^2} - \frac{\partial^2 I}{\partial y^2}$

15	pix_hog_mean	Mean of HOG features in a local window, representing the average level of the gradient distribution	$\text{HOG}_{\text{mean}} = \frac{1}{N} \sum_{i=1}^N \text{HOG}_i$
16	pix_hog_std	Standard deviation of HOG features in a local window, characterizing the degree of dispersion in the gradient distribution	$\text{HOG}_{\text{std}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\text{HOG}_i - \text{HOG}_{\text{mean}})^2}$
17	pix_hog_entropy	Local window HOG feature entropy, which characterizes the complexity of the gradient distribution	$\text{HOG}_{\text{entropy}} = - \sum_{i=1}^N p_i \ln p_i$

Table 2: Spectral Features

2.1 Raw Band Characteristics

No.	Feature Name	Physical meaning	Calculation Formula
1	Blue	Original reflectance in the blue light band	$\text{Blue} = B(x, y)$
2	Green	Raw reflectance in the green wavelength band	$\text{Green} = G(x, y)$
3	Red	Original reflectance in the red wavelength range	$\text{Red} = R(x, y)$
4	NIR	Raw reflectance in the near-infrared band	$\text{NIR} = N(x, y)$

2.2 Local Band Statistical Features

No.	Feature Name	Physical meaning	Calculation Formula
5	Blue_mean	Mean value of the local window in the blue light band	$\text{Blue}_{\text{mean}} = \frac{1}{N} \sum_{i=1}^N B_i$
6	Blue_std	Standard deviation of the local window in the blue light band	$\text{Blue}_{\text{std}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (B_i - \text{Blue}_{\text{mean}})^2}$
7	Blue_skew	Skewness of the local window in the blue band	$\text{Blue}_{\text{skew}} = \frac{\frac{1}{N} \sum_{i=1}^N (B_i - \text{Blue}_{\text{mean}})^3}{\text{Blue}_{\text{std}}^3}$
8	Blue_moment4	Fourth-order central moment of a local window in the blue light band	$\text{Blue}_{\text{moment4}} = \frac{1}{N} \sum_{i=1}^N (B_i - \text{Blue}_{\text{mean}})^4$
9	Green_mean	Mean value of the local window in the green light band	$\text{Green}_{\text{mean}} = \frac{1}{N} \sum_{i=1}^N G_i$

			$Green_{std}$
10	Green_std	Standard deviation of the local window in the green light band	$= \sqrt{\frac{1}{N} \sum_{i=1}^N (G_i - Green_{mean})^2}$
			$Green_{skew}$
11	Green_skew	Local window skew in the green light band	$= \frac{\frac{1}{N} \sum_{i=1}^N (G_i - Green_{mean})^3}{Green_{std}^3}$
			$Green_{moment4}$
12	Green_moment4	Fourth-order central moments in a local window in the green wavelength band	$= \frac{1}{N} \sum_{i=1}^N (G_i - Green_{mean})^4$
			Red_{mean}
13	Red_mean	Mean value of the local window in the red light band	$= \frac{1}{N} \sum_{i=1}^N R_i$
			Red_{std}
14	Red_std	Standard deviation of the local window in the red light band	$= \sqrt{\frac{1}{N} \sum_{i=1}^N (R_i - Red_{mean})^2}$
			Red_{skew}
15	Red_skew	Local window skew in the red light band	$= \frac{\frac{1}{N} \sum_{i=1}^N (R_i - Red_{mean})^3}{Red_{std}^3}$
			$Red_{moment4}$
16	Red_moment4	Fourth-order central moments in a local window in the red light spectrum	$= \frac{1}{N} \sum_{i=1}^N (R_i - Red_{mean})^4$
			NIR_{mean}
17	NIR_mean	Mean of the local window in the near-infrared band	$= \frac{1}{N} \sum_{i=1}^N N_i$
			NIR_{std}
18	NIR_std	Standard deviation of the local window in the near-infrared band	$= \sqrt{\frac{1}{N} \sum_{i=1}^N (N_i - NIR_{mean})^2}$
			NIR_{skew}
19	NIR_skew	Local window skew in the near-infrared band	$= \frac{\frac{1}{N} \sum_{i=1}^N (N_i - NIR_{mean})^3}{NIR_{std}^3}$
			$NIR_{moment4}$
20	NIR_moment4	Fourth-order central moments in a local window of the near-infrared band	$= \frac{1}{N} \sum_{i=1}^N (N_i - NIR_{mean})^4$

2.3 Spectral Index Features

No.	Feature Name	Physical meaning	Calculation Formula
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			ATSAVI
21	ATSAVI	Adjusted Soil-Vegetation Index	$= \frac{1.22 \cdot N - 1.22 \cdot R - 0.03}{1.22 \cdot N + R + 0.08 \cdot (1 + 1.22)}$
22	AVI	Bare Soil Vegetation Index	$AVI = 2 \cdot N - R$
23	ARVI	Atmospheric Impedance Vegetation Index	$ARVI = \frac{N - R - (R - B)}{N + R - (R - B)}$
24	ARVI2	Simplified Atmospheric Impedance Vegetation Index	$ARVI2 = -0.18 + 1.17 \cdot \frac{N - R}{N + R}$
25	BWDRVI	Blue-band Weighted Differential Vegetation Index	$BWDRVI = \frac{0.1 \cdot N - B}{0.1 \cdot N + B}$
26	Chlgreen	Green Light Chlorophyll Index	$Chlgreen = \frac{1}{N/G}$
27	CVI	Chlorophyll Vegetation Index	$CVI = \frac{N \cdot R}{G^2}$
28	CI	Color Index	$CI = \frac{R - B}{R}$
			CTVI
29	CTVI	Corrected Triangular Vegetation Index	$= sign\left(\frac{R - G}{R + G} + 0.5\right) \cdot \sqrt{\left \frac{R - G}{R + G} + 0.5\right }$
30	CRI550	550 nm Carotenoid Reflectance Index	$CRI550 = \frac{1}{B} - \frac{1}{G}$
31	GDVI	Green Band Differential Vegetation Index	$GDVI = N - G$
32	DVIMSS	Corrected Difference Vegetation Index	$DVIMSS = 2.4 \cdot N - R$
33	EVI	Enhanced Vegetation Index	$EVI = \frac{2.5 \cdot (N - R)}{N + 6R - 7.5B + 1}$
34	EVI2	Dual-Band Enhanced Vegetation Index	$EVI_2 = \frac{2.4 \cdot (N - R)}{N + R + 1}$
35	EVI2_2	Second Edition of the Dual-Band Enhanced Vegetation Index	$EVI_{2-2} = \frac{2.5 \cdot (N - R)}{N + 2.4R + 1}$
36	GEMI	Global Environmental Monitoring Index	$n_{gemi} = \frac{2(N^2 - R^2) + 1.5N + 0.5R}{N + R + 0.5}$
37	GARI	Global Atmospheric Impedance Vegetation Index	$GARI = \frac{N - (G - (B - R))}{N - (G + (B - R))}$
38	GLI	Green Leaf Index	$GLI = \frac{2G - R - B}{2G + R + B}$
39	GNDVI	Normalized Difference Vegetation Index (NDVI) in the green band	$GNDVI = \frac{N - G}{N + G}$
40	GOSAVI	Green Band Soil-Adjusted Vegetation Index	$GOSAVI = \frac{N - G}{N + G + 0.16}$

41	GSAVI	Green Band-Corrected Soil-Adjusted Vegetation Index	$GSAVI = \frac{(N - G) \cdot (1 + 0.5)}{N + G + 0.5}$
42	GBNDVI	Normalized Difference Vegetation Index (NDVI) in the green and blue bands	$GBNDVI = \frac{N - (G + B)}{N + (G + B)}$
43	GRNDVI	Normalized Difference Vegetation Index (NDVI) in the green and red bands	$GRNDVI = \frac{N - (G + R)}{N + (G + R)}$
44	Hue	Hue Index	$Hue = \arctan\left(\frac{2R - G - B}{\sqrt{3} \cdot (G - B)}\right)$
45	IVI	Albedo Vegetation Index	$IVI = \frac{N}{R}$
46	IPVI	Permanence Vegetation Index	$IPVI = \frac{N}{N + R} \cdot 2 \left(\frac{R - G}{R + G} + 1 \right)$
47	Intensity	Luminance Intensity Index	$Intensity = \frac{1}{\sqrt{3}} \cdot (R + G + B)$
48	LogR	Near-Infrared/Red Light Logarithmic Ratio Index	$LogR = \ln\left(\frac{N}{R}\right)$
49	mCRIG	Adjusted Green Light Carotenoid Index	$mCRIG = \left(\frac{1}{B} - \frac{1}{G}\right) \cdot N$
50	mCRIRE	Adjusted Beta-Carotene Index	$mCRIRE = \left(\frac{1}{B} - \frac{1}{R}\right) \cdot N$
51	MCARI1	Corrected Chlorophyll Absorption Ratio Index 1	$MCARI1 = 1.2 \cdot [2.5(N - R) - 1.3(N - G)]$
52	MCARI2	Corrected Chlorophyll Absorption Ratio Index 2	$MCARI2 = \frac{1.5 \cdot [2.5(N - R) - 1.3(N - G)]}{\sqrt{(2N + 1)^2 - 6N + 5\sqrt{R} - 0.}}$
53	MSR670	Adjusted Simple Ratio Index	$MSR670 = \frac{N/R - 1}{\sqrt{N/R + 1}}$
54	MSRNir/Red	Near-Infrared - Red Light Correction Ratio Index	$MSRNir/Red = \frac{N/R - 1}{\sqrt{N/R + 1}}$
55	MSAVI	Corrected Soil-Adjusted Vegetation Index	$MSAVI = \frac{2N + 1 - \sqrt{(2N + 1)^2 - 8(N \cdot R)}}{2}$
56	MSAVI _{hyper}	Hyperspectral-Corrected Soil-Adjusted Vegetation Index	$MSAVI_{hyper} = 0.5 \left[2N + 1 - \sqrt{(2N + 1)^2 - 8(N - R)} \right]$
57	MTVI1	Corrected Triangular Vegetation Index 1	$MTVI1 = 1.2 \cdot [1.2(N - G) - 2.5(R - G)]$
58	MTVI2	Corrected Triangular Vegetation Index 2	$MTVI2 = \frac{1.5 \cdot [1.2(N - G) - 2.5(R - G)]}{\sqrt{(2N + 1)^2 - 6N + 5\sqrt{R} - 0.}}$

59	NLI	Nonlinear Vegetation Index	$NLI = \frac{N^2 - R}{N^2 + R}$
60	Norm G	Green Light Normalized Ratio	$\text{Norm_G} = \frac{G}{N + R + G}$
61	Norm NIR	Normalized near-infrared ratio	$\text{Norm_NIR} = \frac{N}{N + R + G}$
62	Norm R	Red Light Normalized Ratio	$\text{Norm_R} = \frac{R}{N + R + G}$
63	PPR	Green-to-Blue Band Ratio Index	$PPR = \frac{G - B}{G + B}$
64	PVR	Green-to-Red Band Ratio Index	$PVR = \frac{G - R}{G + R}$
65	PSNDc2	Normalized Difference Vegetation Index 2	$\text{PSNDc2} = \frac{N - B}{N + B}$
66	PSNDc1	Normalized Difference Vegetation Index 1	$\text{PSNDc1} = \frac{N - B}{N + B}$
67	PSNDb1	Normalized Vegetation Biomass Index 1	$\text{PSNDb1} = \frac{N - R}{N + R}$
68	PSNDa1	Normalized Canopy Index 1	$\text{PSNDa1} = \frac{N - R}{N + R}$
69	NGRDI	Normalized Difference Index of the Green and Red Bands	$\text{NGRDI} = \frac{G - R}{G + R}$
70	BNDVI	Normalized Difference Vegetation Index (NDVI) in the blue band	$\text{BNDVI} = \frac{N - B}{N + B}$
71	GNDVI_74	Normalized Difference Vegetation Index (NDVI) in the green band (repeat)	$\text{GNDVI_74} = \frac{N - G}{N + G}$
72	NDVI	Normalized Vegetation Index	$\text{NDVI} = \frac{N - R}{N + R}$
73	RI	Red-to-Blue Ratio Index	$RI = \frac{R - G}{R + G}$
74	OSAVI	Optimized Soil-Vegetation Index	$\text{OSAVI} = \frac{1.16 \cdot (N - R)}{N + R + 0.16}$
75	RDVI	Rescaled Difference Vegetation Index	$\text{RDVI} = \frac{N - R}{\sqrt{N + R}}$
76	RDVI2	Second Edition of the Renormalized Difference Vegetation Index	$\text{RDVI2} = \frac{N - R}{\sqrt{N + R}}$
77	RBNDVI	Normalized Difference Vegetation Index (NDVI) in the red and blue bands	$\text{RBNDVI} = \frac{N - (R + B)}{N + (R + B)}$
78	RDVI_91	Rescaled Difference Vegetation Index (Repeat)	$\text{RDVI_91} = \frac{N - R}{\sqrt{N + R}}$
79	BGI	Blue-to-Green Band Ratio Index	$\text{BGI} = \frac{B}{G}$
80	BRI	Blue-to-Red Band Ratio Index	$\text{BRI} = \frac{B}{R}$

81	DSWI-4	Drought Monitoring Index 4	$DSWI-4 = \frac{G}{R}$
82	GI	Greenness Index	$GI = \frac{G}{R}$
83	RVI	Ratio Vegetation Index	$RVI = \frac{N}{R}$
84	GRVI	Green-Red Band Ratio Vegetation Index	$GRVI = \frac{N}{G}$
85	IO	Iron Oxide Index	$IO = \frac{R}{B}$
86	RGR	Red-to-Green Band Ratio Index	$RGR = \frac{R}{G}$
87	SRRed/NIR	Red-to-Near-Infrared Ratio Index	$SRRed/NIR = \frac{R}{N}$
88	SAVI	Soil-Vegetation Index	$SAVI = \frac{(N - R) \cdot (1 + 0.5)}{N + R + 0.5}$
89	SARVI	Soil-Atmosphere Impedance Vegetation Index	$SARVI = \frac{(1 + 0.5) \cdot (N - (R - (B - R)))}{N + (R - (B - R)) + 0.5}$
90	SARVI2	Second Edition of the Soil-Atmosphere Impedance Vegetation Index	$SARVI2 = \frac{2.5 \cdot (N - R)}{1 + N + 6R - 7.5B}$
91	SAVI3	Third Edition of the Soil-Vegetation Index	$SAVI3 = \frac{(N - R) \cdot (1 + 0.5)}{N + R + 0.5}$
92	SBL	Soil Background Index	$SBL = N - 2.4R$
93	SQRT(IR/R)	Near-infrared / Red Light Power Ratio Index	$SQRT(IR/R) = \sqrt{\frac{N}{R}}$
94	TNDVI	Triangular Normalized Vegetation Index	$TNDVI = \sqrt{\max\left(\frac{N - R}{N + R}, 0\right) + 0.5}$
95	TSAVI	Conversion-type Soil-Vegetation Index	$TSAVI = \frac{1.22(N - 1.22R - 0.03)}{R + 1.22(N - 0.03) + 0.08(1 - 0.03)}$
96	TSAVI_2	Second Edition of the Convertible Soil-Adjusted Vegetation Index	$TSAVI_2 = \frac{1.22(N - 1.22R - 0.03)}{R + 1.22N - 1.22 \times 0.03}$
97	TVI	Triangular Vegetation Index	$TVI = \sqrt{\max\left(\frac{R - G}{R + G}, 0\right) + 0.5}$
98	VARIGreen	Visible Light Atmospheric Impedance Vegetation Index	$VARIGreen = \frac{G - R}{G + R - B}$
99	WDVI	Weighted Difference Vegetation Index	$WDVI = N - R$

100	WDRVI	Wide Dynamic Range Vegetation Index	$WDRVI = \frac{0.1N - R}{0.1N + R}$
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Table 3: Texture Features

No.	Feature Name	Physical meaning	Calculation Formula
1	glcm_contrast	GLCM contrast, which characterizes the intensity of variations in texture depth	$Contrast = \sum_i \sum_j (i - j)^2 \cdot P(i, j)$
2	glcm_homogeneity	GLCM homogeneity, which characterizes the degree of local uniformity in texture	$Homogeneity = \sum_i \sum_j \frac{P(i, j)}{1 + (i - j)^2}$
3	glcm_energy	GLCM energy, a measure of texture regularity and grayscale uniformity	$Energy = \sum_i \sum_j P(i, j)^2$
4	glcm_entropy	GLCM entropy, which characterizes texture complexity and information content	$Entropy = - \sum_i \sum_j P(i, j) \cdot \log_2 P(i, j)$
5	lbp_u0	Uniform LBP Pattern 0 Histogram Distribution	$lbp_u0 = \frac{N_0}{N_{total}}$
6	lbp_u1	Uniform LBP Pattern 1 Histogram Distribution	$lbp_u1 = \frac{N_1}{N_{total}}$
7	lbp_u2	Uniform LBP Pattern 2 Histogram Distribution	$lbp_u2 = \frac{N_2}{N_{total}}$
8	lbp_u3	Uniform LBP Pattern 3 Histogram Distribution	$lbp_u3 = \frac{N_3}{N_{total}}$
9	lbp_u4	Uniform LBP Pattern 4 Histogram Distribution	$lbp_u4 = \frac{N_4}{N_{total}}$
10	lbp_u5	Uniform LBP Pattern 5 Histogram Distribution	$lbp_u5 = \frac{N_5}{N_{total}}$
11	lbp_u6	Uniform LBP Pattern 6 Histogram Distribution	$lbp_u6 = \frac{N_6}{N_{total}}$
12	lbp_u7	Uniform LBP Pattern 7 Histogram Distribution	$lbp_u7 = \frac{N_7}{N_{total}}$
13	tamura_coarseness	Tamura roughness, a measure of the coarseness of texture particles	$Coarse = \frac{1}{MN} \sum_{x,y} S(x, y)$

			$\mu = \frac{1}{N} \sum_{i=1}^N I_i, \sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (I_i - \mu)^2}$
14	tamura_contrast	Tamura Contrast, characterizing the depth of texture	$m_4 = \frac{1}{N} \sum_{i=1}^N (I_i - \mu)^4, \alpha_4 = \frac{m_4}{\sigma^4}$ $\text{Contrast} = \frac{\sigma}{\sqrt[4]{\alpha_4}}$
15	tamura_directionality	Tamura Directionality, a measure of the consistency of texture direction	$\text{Ent} = - \sum p_k \log_2 p_k$ $\text{Directionality} = 1 - \frac{\text{Ent}}{\log_2 N_{\text{bin}}}$
16	tamura_line likeness	Tamura's linearity index, which characterizes the degree of linear texture	$\text{Linelikeness} = \frac{\mathbb{E}(g_x \cdot g_y)}{\mathbb{E}(\sqrt{g_x^2 + g_y^2})}$
17	tamura_regularity	Tamura regularity, a measure of the regularity of texture repetition	$\text{Regularity} = \frac{1}{\text{std}([\text{Coarse}, \text{Contrast}, \text{Dir}, \text{Line}])}$
18	tamura_roughness	Tamura's Composite Index of Coarseness, Roughness, and Contrast	$\text{Roughness} = \text{Coarseness} + \text{Contrast}$

Table 4 Spatial Relationship Features

No.	Feature Name	Physical meaning	Calculation Formula
1	nearest_dist	Euclidean distance from a single target to the nearest neighbor	$d_{\text{near}} = \min(c_i - c_j _2), i \neq j$
2	mean_dist	The average distance from a single target to all other targets	$d_{\text{mean}} = \frac{1}{M-1} \sum_{j \neq i} c_i - c_j _2$
3	dist_std	The standard deviation of the distances between targets, which characterizes the dispersion of the distribution	σ_d $= \sqrt{\frac{1}{M-1} \sum_{j \neq i} (c_i - c_j _2 - d_{\text{mean}})^2}$
4	dist_max	The distance from a single target to the farthest neighboring target	$d_{\text{max}} = \max(c_i - c_j _2), i \neq j$
5	spatial_compactness	The distance from a target to the global cluster center, which characterizes the cluster's compactness	$D_c = c_i - \bar{c} _2$
6	mean_min_ratio	The ratio of the average distance to the nearest-neighbor distance	$\text{Ratio} = \frac{d_{\text{mean}}}{d_{\text{near}}}$
7	dist_to_nearest_object	Euclidean distance from a single pixel to the center of the nearest target	$d_p = \min(p - c_i _2)$

8	local_object_density	Local target density within the sliding window	$\rho_l = \frac{M_{win}}{Win^2}$
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Table 5: Topographic Features

No.	Feature Name	Physical meaning	Calculation Formula
1	DEM	Original digital elevation values	$DEM = E(x, y)$
2	Slope	Surface slope, which indicates the steepness or gentleness of the terrain	$Slope = S(x, y)$
3	Aspect_sin	Slope sine function, eliminating angular periodicity	$Aspect_sin = \sin(A_{rad})$
4	Aspect_cos	Cosine of the slope, eliminating angular periodicity	$Aspect_cos = \cos(A_{rad})$
5	Ruggedness	Relief, which indicates the degree of surface fragmentation	$Ruggedness = R(x, y)$
6	DEM_local_mean	Mean of the local elevation window	$DEM_{mean} = \frac{1}{N} \sum_{i=1}^N E_i$
7	DEM_local_std	Standard deviation of the local elevation window	$DEM_{std} = \sqrt{\frac{1}{N} \sum_{i=1}^N (E_i - DEM_{mean})^2}$
8	DEM_local_relief	The elevation varies little locally	$DEM_{relief} = \max(E_i) - \min(E_i)$
9	DEM_local_cv	Coefficient of Variation for Elevation	$DEM_{cv} = \frac{DEM_{std}}{DEM_{mean}}$
10	Slope_local_mean	Local window mean of the gradient	$Slope_{mean} = \frac{1}{N} \sum_{i=1}^N S_i$
11	Slope_local_std	Standard deviation of the local gradient window	$Slope_{std} = \sqrt{\frac{1}{N} \sum_{i=1}^N (S_i - Slope_{mean})^2}$
12	Slope_local_relief	The slope varies slightly in some areas	$Slope_{relief} = \max(S_i) - \min(S_i)$
13	Slope_local_cv	Coefficient of variation for slope	$Slope_{cv} = \frac{Slope_{std}}{Slope_{mean}}$
14	Rugged_local_mean	Mean of the local window of undulation	$Rugged_{mean} = \frac{1}{N} \sum_{i=1}^N R_i$
15	Rugged_local_std	Standard deviation of the local window of undulation	$Rugged_{std} = \sqrt{\frac{1}{N} \sum_{i=1}^N (R_i - Rugged_{mean})^2}$

16	Rugged_local_relief	Local variation in undulation	$Rugged_{relief} = \max(R_i) - \min(R_i)$
17	Rugged_local_cv	Coefficient of Variation of Undulation	$Rugged_{cv} = \frac{Rugged_{std}}{Rugged_{mean}}$
18	Aspect_resultant	Aspect vector, characterizing local aspect consistency	$Aspect_{res} = \sqrt{(\sin A)^2 + (\cos A)^2}$
19	Aspect_entropy	Slope entropy, which characterizes the degree of local slope disorder	$Aspect_{ent} = - \sum p_i \ln p_i$
20	DEM_grad_magnitude	Elevation gradient amplitude, which characterizes the degree of topographic variation	$E_{grad} = \sqrt{\left(\frac{\partial E}{\partial x}\right)^2 + \left(\frac{\partial E}{\partial y}\right)^2}$
21	DEM_laplacian	Elevation Laplace, characterizing areas of terrain undulation and abrupt changes	$E_{lap} = \frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2}$
22	Slope_grad	Slope gradient amplitude, which characterizes the intensity of slope variation	$S_{grad} = \sqrt{\left(\frac{\partial S}{\partial x}\right)^2 + \left(\frac{\partial S}{\partial y}\right)^2}$
23	Rugged_grad	Relief gradient amplitude, characterizing variations in surface fragmentation	$R_{grad} = \sqrt{\left(\frac{\partial R}{\partial x}\right)^2 + \left(\frac{\partial R}{\partial y}\right)^2}$

Table 6: InSAR Subsidence Characteristics

No.	Feature Name	Physical meaning	Calculation Formula
1	CJ	Original surface settlement rate	$CJ = C(x, y)$
2	SULV	Auxiliary settlement monitoring values	$SULV = S(x, y)$
3	CJ_LocalMean	CJ Local Window Mean	$CJ_{mean} = \frac{1}{N} \sum_{i=1}^N C_i$
4	CJ_LocalStd	CJ Local Window Standard Deviation	$CJ_{std} = \sqrt{\frac{1}{N} \sum_{i=1}^N (C_i - CJ_{mean})^2}$
5	SULV_LocalMean	SULV (Local Window Mean)	$SULV_{mean} = \frac{1}{N} \sum_{i=1}^N S_i$
6	SULV_LocalStd	SULV (Standard Deviation of Local Windows)	$SULV_{std} = \sqrt{\frac{1}{N} \sum_{i=1}^N (S_i - SULV_{mean})^2}$
7	CJ_LocalAnomaly	CJ Local Deformation Outliers	$CJ_{ano} = C(x, y) - CJ_{mean}$

8	SULV_LocalAnomaly	SULV Local Deformation Anomalies	$\text{SULV}_{\text{ano}} = S(x, y) - \text{SULV}_{\text{mean}}$
9	CJ_GradX	First-order gradient in the X direction of the sedimentation field	$C_x = \frac{\partial C(x, y)}{\partial x}$
10	CJ_GradY	First-order gradient in the Y direction of the sedimentation field	$C_y = \frac{\partial C(x, y)}{\partial y}$
11	CJ_GradMag	The magnitude of the settlement gradient, which characterizes the severity of deformation	$C_{\text{mag}} = \sqrt{C_x^2 + C_y^2}$
12	CJ_GradOri	The direction of the settlement gradient, which characterizes the direction of deformation extension	$C_{\text{ori}} = \arctan 2(C_y, C_x)$
13	CJ_Laplacian	Laplace's field of settlement, characterizing second-order deformation discontinuities	$C_{\text{lap}} = \frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2}$
14	CJ_Curvature	Curvature of the settlement field, characterizing the topography of deformation	$C_{\text{urv}} = \frac{\partial^2 C}{\partial x^2} \cdot \frac{\partial^2 C}{\partial y^2} - \left(\frac{\partial^2 C}{\partial x \partial y} \right)^2$
15	CJ_ActiveMask	Active Sedimentation Binary Mask (Threshold - 0.05)	$\text{Mask} = \begin{cases} 1, C(x, y) < -0.05 \\ 0, C(x, y) \geq -0.05 \end{cases}$
16	ActiveSubsidence_LocalRatio	Percentage of active settling pixels in a local window	$\text{Ratio} = \frac{N_{\text{mask}=1}}{N}$